Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. $(2,-1)\cdot(1,2)$?
2. Does the point $(1,2)$ satisfy $2x-y=0$?
3. Solve $x+y=5$, $x-y=1$.
4. Distance from $(1,2)$ to $(4,6)$?
5. Complete x^2-4x.
6. Solve $x^2+y^2=0$ in $\mathbb{R}^2$.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present the two conditions that characterise an orthogonal projection. Give a geometric proof and an algebraic proof of the perpendicular-bisector property.

Investigate before formalising

Starting from $P(4,3)$ and $d : y = 2x$, compare the distances to three points of d, then find and prove which is the shortest.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.